

NPTEL ETHICAL HACKING & NETWORK COMMUNICATIONS

WEEK 2 DETAILED CLASSROOM STUDY NOTES & EXAM PREPARATION NOTEBOOK
Coverage: Lectures 06 to 10 (IP Fragmentation, Addressing, TCP/UDP, Subnetting, VLSM & CIDR)

MODULE 1: LECTURE 06 — IP ADDRESSING & ROUTING (PART I: PACKET FRAGMENTATION)

1.1 Fundamental Concept: What is IP Fragmentation?

In a global internetwork, data packets traverse diverse physical networks (such as Ethernet, Wi-Fi, optical fiber, and satellite links). Each physical network technology has a strict hardware limit on the maximum size of a frame payload it can carry without splitting it apart. This maximum payload size at the Data Link Layer is called the **Maximum Transfer Unit (MTU)**.

What it is: IP Fragmentation is the process of breaking down a large IP datagram into smaller, compliant pieces (called *fragments*) when it encounters a network segment with an MTU smaller than the packet's total size.

Why it is needed: Without fragmentation, if a 1500-byte packet arrives at a router interface connected to a network with an MTU of 620 bytes, the router would be forced to drop the packet entirely, causing complete communication failure.

How it works step-by-step:

- **Step 1 (Size Check):** When a router receives an IP packet, it compares the packet's Total Length with the MTU of the outgoing network interface.
- **Step 2 (Flag Verification):** The router checks the *Don't Fragment (DF)* bit in the IP header. If $DF = 1$ and $\text{packet size} > \text{MTU}$, the router drops the packet and sends an ICMP 'Destination Unreachable - Fragmentation Needed' error back to the source host.
- **Step 3 (Payload Splitting):** If $DF = 0$, the router divides the original data payload into chunks that fit within the new MTU. Each chunk size (except the last) **MUST** be a multiple of 8 bytes.
- **Step 4 (Header Duplication & Field Updating):** The router attaches a new 20-byte IP header to each fragment, copies the original *Identification (ID)* field, calculates the new *Fragment Offset*, updates the *More Fragments (MF)* flag, and recalculates the Header Checksum.

Technical Term	Technical Definition	Security & System Context
IP Fragmentation	Process of breaking a large IP packet into smaller fragments to traverse links with smaller MTU.	Enables heterogeneous networking; target for fragmentation evasion attacks.
Reassembly	Process of combining received fragments back into the original complete IP datagram.	Occurs only at the destination host in IP (non-transparent mode).
MTU	Maximum Transfer Unit: Maximum size in bytes of a network-layer datagram a link can carry.	Standard Ethernet MTU is 1500 bytes.
Transparent Frag.	Fragmentation and reassembly occur internally within a single intermediate network.	Invisible to downstream networks; creates bottlenecks at exit routers.
Non-Transparent Frag.	Fragments are NOT reassembled by routers; forwarded independently to final host.	Standard method used by IP; parallel routing but higher header overhead.
Identification (ID)	16-bit field in IP header that uniquely identifies the original datagram.	All fragments from the same packet share the exact same ID value.
Fragment Offset	13-bit field specifying position of fragment data relative to original datagram.	Measured in multiples of 8 bytes (64 bits).
Flags (D and M)	3-bit field: Bit 1 = D (Do Not Fragment), Bit 2 = M (More Fragments).	D=1 prevents fragmentation; M=1 indicates more fragments follow, M=0 means last fragment.

1.2 Transparent vs. Non-Transparent Fragmentation

There are two fundamental strategies for managing packet fragmentation across multiple intermediate networks:

1. Transparent Fragmentation: An ingress router fragments a packet entering a small MTU network. These fragments traverse the internal network to a designated *Exit Router (\$R_E\$)*. The exit router collects ALL fragments, reassembles them back into the original unfragmented packet, and forwards it to the next network. To downstream networks, fragmentation is completely invisible ('transparent').

Drawbacks: Overloads the exit router (bottleneck), requires all fragments to route through the exact same exit router, and causes repeated fragmentation/reassembly cycles if traversing multiple small MTU networks.

2. Non-Transparent Fragmentation (The IP Protocol Standard): When a router fragments a packet, it forwards each fragment independently as an autonomous IP packet. Intermediate routers perform NO reassembly. Reassembly is performed solely by the **final destination host**.

Advantages: Allows fragments to take parallel routes through different routers, maximizing throughput and relieving intermediate routers of reassembly buffer memory overhead.

Drawbacks: Significant header overhead (each fragment requires its own 20-byte IP header) and increased CPU load on the destination host.

COMMON CONFUSION / MISTAKE

WHY IS FRAGMENT OFFSET MEASURED IN MULTIPLES OF 8 BYTES?

The Fragment Offset field in the IP header is only 13 bits wide. If it counted individual bytes, $2^{13} = 8192$ bytes would be the maximum packet offset possible. By scaling the offset in multiples of 8 bytes ($8 \times 2^{13} = 65,536$ bytes), the 13-bit field can represent any offset up to the maximum theoretical IP packet size of 65,535 bytes!

1.3 Step-by-Step Numerical Solved Example (From Lecture)

Let us walk through the exact step-by-step fragmentation example presented in NPTEL Lecture 06 to master the mathematical calculations:

Problem Scenario: An original IP packet has **1000 bytes of data** and a **20-byte IP header** (Total Size = 1020 bytes). Parameters: Identification ID = 5, DF = 0, MF = 0, Fragment Offset = 0.

Phase 1: Packet enters Network 1 with MTU = 620 bytes.

- Maximum data payload per fragment = $620 - 20 = 600$ bytes (which is divisible by 8: $600 / 8 = 75$).
- **Fragment 1:** Data = 600B, Header = 20B (Total 620B), ID = 5, Offset = 0, **MF = 1**.
- **Fragment 2:** Data = $1000 - 600 = 400$ B, Header = 20B (Total 420B), ID = 5, **Offset = $600 / 8 = 75$** , **MF = 0**.

Phase 2: Fragments enter Network 2 with MTU = 400 bytes.

- Max data capacity = $400 - 20 = 380$ B. Largest multiple of 8 ≤ 380 is **376 bytes**.
- *Fragment 1 (620B) is split into two sub-fragments:*
 - **Frag 1.1:** Data = 376B, Header = 20B (Total 396B), ID = 5, Offset = 0, **MF = 1**.
 - **Frag 1.2:** Data = $600 - 376 = 224$ B, Header = 20B (Total 244B), ID = 5, **Offset = $376 / 8 = 47$** , **MF = 0** (last part of original Frag 1).
- *Fragment 2 (420B) is split into two sub-fragments:*
 - **Frag 2.1:** Data = 376B, Header = 20B (Total 396B), ID = 5, **Offset = 75** (inherits original offset), **MF = 1**.
 - **Frag 2.2:** Data = $400 - 376 = 24$ B, Header = 20B (Total 44B), ID = 5, **Offset = $(75 * 8 + 376) / 8 = 976 / 8 = 122$** , **MF = 0**.

Total Transmitted Bytes across Net 2: $396 + 244 + 396 + 44 = 1080$ bytes (80 bytes of extra header overhead due to non-transparent fragmentation).

1.4 Ethical Hacking Relevance & Security Impact

IP fragmentation mechanics introduce several classical security vulnerabilities that ethical hackers and penetration testers must evaluate:

- **Teardrop Attack (Overlapping Offsets):** Attackers craft malicious IP fragments with overlapping offset fields (e.g., Frag 1 offset 0 to 600, Frag 2 offset 300 to 700). When vulnerable OS kernels attempt to reassemble these overlapping blocks, buffer allocation routines fail, leading to system crashes or Denial of Service (DoS).
- **NIDS / Firewall Evasion:** Intrusion Detection Systems inspect packet payloads for attack signatures (e.g., SQL injection keywords). Attackers deliberately split an attack signature across two fragments (e.g., 'SE' in Frag 1, 'LECT' in Frag 2). If the Firewall/NIDS does not reassemble fragments before inspection, the signature bypasses detection.

- **Fragment Flooding DoS:** Attackers flood a target host with thousands of incomplete IP fragments with MF = 1, omitting the final fragment. The target host holds reassembly buffers open in memory until time-out limits expire, consuming RAM and denying resources to legitimate traffic.

EXAM POINT

NPTEL EXAM ALERT: IP MTU VS ETHERNET MTU

Although the theoretical maximum size of an IP datagram is 65,535 bytes, Ethernet data link layers limit frames to 1500 bytes. To prevent constant link-layer fragmentation, standard IP operating systems cap their practical IP MTU to 1500 bytes!

MODULE 2: LECTURE 07 — IP ADDRESSING & ROUTING (PART II: CLASSFUL ADDRESSING)

2.1 Fundamental Concept: The Structure of an IP Address

Every device connected to the Internet must possess a globally unique address to ensure data packets reach the intended destination without ambiguity. In IPv4, an **IP Address is a 32-bit binary number**, offering a theoretical total address space of 2^{32} ■ approx 4.29\$ billion distinct addresses.

Dotted Decimal Notation: Because a 32-bit binary string (e.g., `01000010 10000110 00110000 01111110`) is impossible for humans to memorize, it is formatted into four 8-bit octets separated by dots. Each octet is converted to its decimal value (range 0 to 255). For example, `66.134.48.126`.

Hierarchical Structure (Network ID + Host ID): Just as a postal address splits into [City/Zip Code] and [Street/House Number], every IP address is divided into two parts:

1. **Network Number (Net ID):** Identifies the specific network to which the host belongs. Routers use the Net ID to forward traffic across the Internet.
2. **Host Number (Host ID):** Uniquely identifies a specific device interface within that local network.

2.2 Classful Addressing Architecture

Historically, the Internet address space was divided into 5 classes (Class A, B, C, D, and E) defined by the leading bits of the first octet:

Class	Leading Bits	Network / Host Split	First Octet Range	Max Usable Networks	Usable Hosts per Net
Class A	0...	8 bits Net / 24 bits Host	1.0.0.0 to 126.255.255.255	126 (0 & 127 reserved)	16,777,214 ($2^{24}-2$)
Class B	10...	16 bits Net / 16 bits Host	128.0.0.0 to 191.255.255.255	16,384 (2^{14})	65,534 ($2^{16}-2$)
Class C	110...	24 bits Net / 8 bits Host	192.0.0.0 to 223.255.255.255	2,097,152 (2^{21})	254 (2^8-2)
Class D	1110...	Multicast Group Addressing	224.0.0.0 to 239.255.255.255	N/A (Group Traffic)	N/A
Class E	1111...	Reserved for Experimental Use	240.0.0.0 to 255.255.255.255	N/A	N/A

2.3 Special Host Conventions & Reserved Addresses

Within any IP network, two specific host address combinations are reserved and **cannot be assigned to individual computers**:

- **All 0s in Host Part (Network Address):** Represents the network itself. For example, in `118.0.0.0/8`, the host part is all 0s. Routers store network addresses in their routing tables to direct traffic.
- **All 1s in Host Part (Directed Broadcast Address):** Used to send a packet to ALL hosts within that specific network. For example, `118.255.255.255` broadcasts to every host in network `118.0.0.0`.

KEY CONCEPT TO REMEMBER

WHY SUBTRACT 2 FROM TOTAL HOST COMBINATIONS? For \$h\$ host bits, the total theoretical bit combinations is 2^h . However, Usable Hosts = $2^h - 2$ because we MUST subtract 2 reserved addresses: (1) The Network Address (all host bits 0) and (2) The Broadcast Address (all host bits 1).

2.4 Special Purpose IP Address Ranges

Certain IP address blocks are designated for specialized system roles and private internal use:

- **Private IP Ranges (RFC 1918):** Reserved for use within private corporate or home networks. They are **non-routable on the public Internet**. Routers drop packets containing private destination IPs on the Internet.
 - *Class A Private:* `10.0.0.0` to `10.255.255.255` (`10.0.0.0/8`)
 - *Class B Private:* `172.16.0.0` to `172.31.255.255` (`172.16.0.0/12` - 16 contiguous Class B blocks)
 - *Class C Private:* `192.168.0.0` to `192.168.255.255` (`192.168.0.0/16` - 256 contiguous Class C blocks)*Network Address Translation (NAT)* translates private internal IPs to a public IP when connecting to the Internet.
- **Loopback Address Range:** `127.0.0.0` to `127.255.255.255` (commonly `127.0.0.1` / `localhost`). Used by an operating system to send network traffic back to itself for internal process testing and local service binding. Loopback packets never touch physical network hardware.
- **Default Network Address:** `0.0.0.0` represents an unassigned address or default internet route placeholder.
- **Limited Broadcast Address:** `255.255.255.255` broadcasts a message to every device on the current physical LAN segment. Routers block limited broadcast packets to prevent network-wide broadcast storms.

2.5 Ethical Hacking Relevance & Security Impact

- **Network Reconnaissance:** Ethical hackers scan target IP ranges to map live hosts. Identifying whether an IP is public vs private helps determine if the target host is directly internet-facing or behind a NAT firewall.
- **IP Spoofing & Ingress Filtering:** Malicious actors spoof public IP addresses to bypass IP-based access controls. Modern networks employ BCP 38 / ingress filtering on border routers to block outbound packets whose source IP does not match the internal private subnet.
- **Loopback Service Binding Misconfiguration:** Developers often bind sensitive management services (e.g., Redis, internal admin APIs) to `0.0.0.0` (all interfaces) instead of `127.0.0.1` (loopback only), unintentionally exposing internal databases to external network attackers.

MODULE 3: LECTURE 08 — TRANSPORT LAYER: TCP & UDP (PART I)

3.1 Role of the Transport Layer: Process-to-Process Communication

While the Internet Protocol (IP) at the Network Layer delivers packets from one machine to another (**host-to-host delivery**), a single host may run dozens of applications simultaneously (web browser, email client, video call, SSH session). The primary job of the **Transport Layer** is to provide **process-to-process delivery** — ensuring incoming network data reaches the exact application process intended.

3.2 TCP vs. UDP Architectural Comparison

Applications pass data to one of two fundamental transport protocols depending on whether they prioritize *absolute reliability* or *high speed & low latency*:

Protocol Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Communication Model	Connection-Oriented (3-way handshake required)	Connectionless (No connection setup)
Reliability Level	100% Reliable (ACKs, Retransmissions, Checksums)	Unreliable / Best-Effort (No ACKs, No Retransmit)
Data Format	Byte-Stream Service (Continuous stream of bytes)	Datagram Service (Independent packet messages)
Delivery Order	Strictly Ordered & Sequenced (Reordered by TCP)	Unordered (Packets arrive in arbitrary order)
Flow & Congestion Control	Yes (Dynamic sliding window flow control)	None (Sends data as fast as application provides)
Header Overhead	20 to 60 bytes header	Fixed 8 bytes header
Primary Use Cases	Web (HTTP/HTTPS), Email (SMTP), File Transfer (FTP), SSH	DNS queries, Video/Audio Streaming, Gaming, TFTP

3.3 Core Reliability Mechanisms of TCP

1. **Checksum:** Every TCP segment contains a 16-bit checksum covering the header, data, and an IP pseudo-header. Corrupted segments are silently discarded.
2. **Positive Acknowledgements (ACK):** The receiver explicitly sends ACK packets back to the sender containing the sequence number of the next byte expected.
3. **Timeouts & Retransmission:** The sender starts a Retransmission Timer for every sent segment. If no ACK arrives before timeout, the segment is automatically retransmitted.
4. **End-to-End Flow Control:** The receiver advertises its available buffer space (Window Size) in every ACK header, preventing a fast sender from flooding a slow receiver.
5. **Sequencing:** TCP assigns a 32-bit Sequence Number to every byte of data. Out-of-order IP datagrams are held in receiver buffers and reassembled into exact sequence before delivery to the application.

3.4 Port Numbers & Service File Mapping

To direct data to specific running processes, the Transport Layer uses **16-bit Port Numbers** (allowing $2^{16} = 65,536$ unique ports, numbered 0 to 65535):

- **Well-Known Ports (0 to 1023, extended to 4095):** Standardized ports reserved for well-known server services:
 - Port 20 & 21: FTP (File Transfer Protocol - Data & Control)
 - Port 22: SSH (Secure Shell) | Port 23: Telnet
 - Port 25: SMTP (Simple Mail Transfer Protocol)
 - Port 53: DNS (Domain Name System - TCP & UDP)
 - Port 69: TFTP (Trivial File Transfer Protocol - UDP)
 - Port 80: HTTP | Port 443: HTTPS
- **Ephemeral Ports (> 4095):** Temporary, dynamic ports assigned automatically by the operating system to client programs (e.g., your web browser) when initiating outbound connections.
- **Operating System Services File:** Mappings between service names, port numbers, and protocols are stored in text system files:
 - Linux/Unix: `/etc/services`
 - Windows: `C:\Windows\System32\drivers\etc\services`

3.5 TCP Association (The 5-Tuple)

A full TCP connection stream is uniquely identified across the entire Internet by a 5-element structure called an **Association (5-Tuple)**:

```
Association = < Protocol (TCP), Local IP, Remote IP, Local Port, Remote Port >
```

EXAM POINT

EXAM ALERT: WELL-KNOWN PORTS ON EXAMS

Remember these key port numbers for NPTEL exams: Echo = Port 7, Daytime = Port 13, FTP Control = Port 21, SSH = Port 22, Telnet = Port 23, SMTP = Port 25, DNS = Port 53, TFTP = Port 69, HTTP = Port 80.

MODULE 4: LECTURE 09 — TRANSPORT LAYER: TCP & UDP (PART II)

4.1 TCP Header Structure Breakdown (20 to 60 Bytes)

A TCP segment consists of a TCP Header followed by application data bytes. The TCP header length ranges from a minimum of **20 bytes** (without options) up to **60 bytes** (with max options):

Field Name	Size (Bits)	Functional Description
Source Port	16 bits	Identifies the sending application process.
Destination Port	16 bits	Identifies the receiving application process on destination host.
Sequence Number	32 bits	Specifies the byte sequence number of the first data byte in this segment.
Acknowledgement Number	32 bits	Specifies the next byte sequence number the receiver expects to receive.

Field Name	Size (Bits)	Functional Description
Header Length (Data Offset)	4 bits	Specifies header size in 32-bit words (min value 5 = 20 bytes, max 15 = 60 bytes).
Reserved	6 bits	Reserved bits set to 0.
Control Flags	6 bits	URG, ACK, PSH, RST, SYN, FIN control flags managing connection lifecycle.
Window Size	16 bits	Advertises current available receiver buffer space (Flow Control).
Checksum	16 bits	Mandatory error checking over header, data payload, and IP pseudo-header.
Urgent Pointer	16 bits	If URG=1, points to the byte offset where urgent data ends.
Options & Padding	0 to 40 bytes	Optional parameters such as Maximum Segment Size (MSS) and Window Scaling.

4.2 TCP Control Flags Explained

- **SYN (Synchronize):** Set to 1 during the 3-way handshake to initiate a connection and synchronize sequence numbers.
- **ACK (Acknowledge):** Set to 1 to indicate that the Acknowledgement Number field contains valid sequence data.
- **FIN (Finish):** Set to 1 to request graceful termination of a TCP connection stream.
- **RST (Reset):** Set to 1 to instantly abort and reset an abnormal or rejected connection.
- **PSH (Push):** Set to 1 to instruct the receiver to push buffered data immediately to the application without waiting for buffer filling.
- **URG (Urgent):** Set to 1 to signal that the segment contains high-priority urgent data.

4.3 TCP Connection Lifecycle: Handshake Protocols

A. TCP 3-Way Handshake (Connection Establishment):

Before exchanging application data, client and server synchronize initial sequence numbers:

1. **Step 1 (SYN):** Client $\rightarrow$ Server sending `[SYN, Seq = x]`. Client enters `SYN\_SENT` state.
2. **Step 2 (SYN-ACK):** Server $\rightarrow$ Client sending `[SYN, ACK, Seq = y, Ack = x + 1]`. Server allocates Transmission Control Block (TCB) memory and enters `SYN\_RCVD` state.
3. **Step 3 (ACK):** Client $\rightarrow$ Server sending `[ACK, Seq = x + 1, Ack = y + 1]`. Both enter `ESTABLISHED` state.

B. TCP 4-Way Handshake (Connection Termination):

Gracefully closes connection in both directions:

1. **Step 1 (FIN):** Client $\rightarrow$ Server sending `[FIN, Seq = u]`. Client enters `FIN\_WAIT\_1`.
2. **Step 2 (ACK):** Server $\rightarrow$ Client sending `[ACK, Ack = u + 1]`. Server enters `CLOSE\_WAIT`; Client enters `FIN\_WAIT\_2`. (Half-close: Server can still send remaining data).
3. **Step 3 (FIN):** Server $\rightarrow$ Client sending `[FIN, Seq = v]`. Server enters `LAST\_ACK`.
4. **Step 4 (ACK):** Client $\rightarrow$ Server sending `[ACK, Ack = v + 1]`. Client enters `TIME\_WAIT` (waits timer $2 \times \text{ext}\{\text{MSL}\}$ before full close).

4.4 UDP Header Format (Fixed 8 Bytes)

In stark contrast to TCP, UDP has an extremely simple fixed **8-byte header** comprising only 4 fields:

```
`[ Source Port (16b) | Destination Port (16b) | Length (16b) | Checksum (16b) ]`
```

4.5 Security Vulnerability: Half-Open Connections & SYN Flood DoS Attack

The Vulnerability: In Step 2 of the 3-Way Handshake, when the server receives a `SYN` request, it allocates TCB memory resources and holds the connection in a *half-open state* (`SYN\_RCVD`) while waiting for the client's final `ACK`.

The SYN Flood Attack Mechanism:

1. The attacker sends thousands of spoofed `SYN` packets to a server with fake/unreachable source IP addresses.
2. The server allocates TCB memory for each `SYN`, sends `SYN-ACK`s to the fake IPs, and waits for final `ACK`s.
3. Because the source IPs are fake, the `SYN-ACK`s are lost and final `ACK`s **never arrive**.
4. The server's connection backlog queue (SYN Queue) becomes completely full. When legitimate users attempt to connect, the server rejects their `SYN` requests, resulting in a **Denial of Service (DoS)**.

Countermeasures: SYN Cookies (cryptographic sequence numbers avoiding TCB allocation until final ACK), reducing SYN timeout limits, and firewall SYN Proxies.

KEY CONCEPT TO REMEMBER

HALF-OPEN STATE DEFINITION

A half-open connection is a state where a server has allocated TCB memory and responded with a SYN-ACK, but the client has not completed the 3-way handshake with a final ACK.

MODULE 5: LECTURE 10 — IP SUBNETTING: MASKS, VLSM & CIDR

5.1 Fundamentals of IP Subnetting & Network Masks

What it is: IP Subnetting is the process of borrowing bits from the host portion of an IP address to divide a large network into smaller, independent sub-networks (subnets).

Why it is needed: In a flat network with 65,000 hosts, a single broadcast packet causes broadcast storms across all devices. Subnetting isolates traffic, reduces broadcast domains, improves security, and optimizes address space.

Bitwise ANDing with Subnet Mask: A 32-bit Subnet Mask contains 1s in network/subnet bit positions and 0s in host positions. Performing a bitwise `AND` between an IP address and its mask reveals the Subnet Network ID:

```
`IP Address: 144.016.072.020 -> 10010000.00010000.00000000.00010100`
`Subnet Mask: 255.255.192.000 -> 11111111.11111111.11000000.00000000`
`-----`
`Subnet ID: 144.016.064.000 -> 10010000.00010000.01000000.00000000`
```

5.2 Variable Length Subnet Mask (VLSM)

Problem with Fixed Subnetting: Fixed-size subnetting forces all subnets to have equal sizes. If Dept 1 needs 110 hosts and Dept 2 needs 10 hosts, a fixed `/25` mask gives 126 hosts to both, wasting 116 addresses in Dept 2.

VLSM Solution: VLSM allows subnets of *different* sizes within the same network space by applying different custom masks based on specific host requirements.

Step-by-Step VLSM Worked Example (Class C 192.203.17.0/24):

- Requirements: Dept 1 (110 hosts), Dept 2 (45 hosts), Dept 3 (50 hosts).
- **Step 1 (Largest Dept):** Dept 1 needs 110 hosts $\implies 2^7 - 2 = 126$ usable hosts. Borrow 1 bit ($\implies /25$).
 - Subnet 1 (Dept 1): `192.203.17.0/25` (Mask `255.255.255.128`). Range: `192.203.17.0` to `192.203.17.127`. Assigned to Dept 1.
- **Step 2 (Remaining Subnets):** Remaining block is `192.203.17.128/25`. Subdivide it for Dept 2 (45 hosts) and Dept 3 (50 hosts) $\implies 2^6 - 2 = 62$ usable hosts. Borrow 1 more bit ($\implies /26$).
 - Subnet 2.1 (Dept 2): `192.203.17.128/26` (Mask `255.255.255.192`). Range: `.128` to `.191`. Assigned to Dept 2.
 - Subnet 2.2 (Dept 3): `192.203.17.192/26` (Mask `255.255.255.192`). Range: `.192` to `.255`. Assigned to Dept 3.

5.3 Classless Inter-Domain Routing (CIDR)

CIDR completely eliminates rigid Class A, B, and C boundaries. An IP block is specified using **slash notation** (`/M`), where `M` is the network prefix length in bits:

- Prefix Length `M` = Network bits. Host bits `H` = $32 - M$.
- Total Addresses in Block = $2^H = 2^{32-M}$. Usable Hosts = $2^{32-M} - 2$.
- **Rule 1:** The number of addresses in any CIDR block MUST be a power of 2 (2, 4, 8, 16, 32, ...).
- **Rule 2:** The starting address of a CIDR block MUST be divisible by the total block size (host bits in starting address must be all 0s).

5.4 Step-by-Step CIDR Numerical Analysis (From Lecture)

Given CIDR Block: ``144.16.192.40/29``

- Network Prefix $M = 29$ bits. Host bits $H = 32 - 29 = 3$ bits.
- Total Block Size = $2^3 = 8$ addresses.
- **Divisibility Check:** Binary of 40 = ``00101000``. The last 3 bits are ``000`` $\implies$ 40 is divisible by 8 (Valid Start Address!).
- **Start Network Address:** ``144.16.192.40`` (Host bits set to ``000``).
- **End Broadcast Address:** ``144.16.192.47`` (Host bits set to ``111``, binary ``00101111`` = 47).
- **Usable Host IP Range:** ``144.16.192.41`` to ``144.16.192.46`` (6 usable hosts).

5.5 Ethical Hacking Relevance & Security Impact

- **CIDR Target Scanning:** Ethical hackers specify target subnets using CIDR notation in scanning tools like Nmap (e.g., ``nmap -sV 192.168.1.0/24``).
- **Subnet Discovery & Pivoting:** Identifying secondary subnets via routing tables on compromised hosts allows attackers to pivot deeper into isolated internal networks.

MODULE 6: WEEK 2 REVISION & NPTEL EXAM PREPARATION PAGES

6.1 Quick Week 2 Core Revision Summary

- **IP Fragmentation:** Occurs when IP packet size > link MTU. Non-transparent fragmentation is IP standard — fragments are routed independently and reassembled ONLY at destination host.
- **Fragmentation Header Fields:** ID (16-bit, unique packet ID), Flags (DF=1 prevents frag, MF=1 indicates more frags follow), Fragment Offset (13-bit, measured in 8-byte multiples).
- **IP Addressing:** 32-bit dotted decimal notation. Net ID + Host ID. Class A (0), Class B (10), Class C (110), Class D (Multicast 1110), Class E (Reserved 1111).
- **Special Addresses:** Network Address (host bits all 0), Broadcast Address (host bits all 1), Private Ranges (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16), Loopback (127.0.0.1).
- **Transport Layer Protocols:** TCP (connection-oriented, reliable, byte-stream, 20-60B header) vs UDP (connectionless, unreliable, datagram, 8B header).
- **Port Numbers:** 16-bit (0-65535). Well-Known (0-1023/4095) vs Ephemeral (>4095). Mapped in /etc/services.
- **TCP Lifecycle:** 3-Way Handshake (SYN → SYN-ACK → ACK), 4-Way Handshake (FIN → ACK → FIN → ACK).
- **SYN Flood DoS Attack:** Exploits half-open TCP state (SYN_RCVD) by sending flooded SYN requests with spoofed IPs, exhausting server TCB queue.
- **Subnetting & VLSM:** Borrowing host bits for subnetting. Bitwise IP AND Mask = Subnet ID. VLSM permits custom subnets of varying sizes.
- **CIDR Notation:** IP/Prefix (/M). Host bits \$H = 32 - M\$. Block size = \$2^H\$. Usable hosts = \$2^H - 2\$. Starting address must be divisible by block size.

6.2 Master Formula Sheet

Formula Name	Mathematical Equation	Variables & Description
Fragment Offset	Offset = Preceding Data Bytes / 8	Calculates 13-bit offset field value (must be integer).
Max Fragment Payload	Payload = MTU - IP Header Size	Max data payload per fragment (rounded down to multiple of 8).
Usable Hosts / Subnet	Usable Hosts = $2^H - 2 = 2^{(32-M)} - 2$	H = Host bits, M = Prefix length (Subtract 2 for Net & Broadcast).
CIDR Block Size	Block Size = $2^H = 2^{(32-M)}$	Total IP addresses in CIDR block (must be power of 2).
Bitwise Subnet ID	Subnet ID = IP Address AND Subnet Mask	Bitwise AND operation on binary representations.
TCP Header Bytes	Header Bytes = Data Offset Field Value * 4	Converts 4-bit Data Offset value (words) into header byte count.

6.3 Important Master Comparisons

Comparison Axis	Option / Technique A	Option / Technique B
Fragmentation Type	Transparent: Reassembled at exit router of intermediate network.	Non-Transparent (IP Standard): Reassembled ONLY at destination host.
Addressing System	Classful: Fixed Class A (/8), Class B (/16), Class C (/24) boundaries.	CIDR: Flexible prefix length (/M) removing class constraints.
Transport Protocol	TCP: Connection-oriented, reliable, byte-stream, flow-controlled.	UDP: Connectionless, unreliable, datagram, lightweight.
Handshake Type	3-Way Handshake: SYN → SYN-ACK → ACK (Establishes connection).	4-Way Handshake: FIN → ACK → FIN → ACK (Terminates connection).
Subnetting Scheme	Fixed-Size Subnetting: Equal subnet sizes across network.	VLSM: Variable subnet sizes customized per department.

6.4 Important Solved Examples

SOLVED EXAMPLE 1: Fragment Offset Calculation

Problem: A router receives an IP packet carrying 1480 bytes of data payload. It fragments it into two pieces for a link with MTU = 800 bytes. Calculate the Fragment Offset for the second fragment.

Solution:

1. Max payload for Fragment 1 = $\text{ext}\{\text{MTU}\} - 20 = 800 - 20 = 780$ bytes.
2. Largest multiple of 8 ≤ 780 is 776 bytes ($776 / 8 = 97$).
3. Fragment 1 carries 776 bytes of data payload.
4. Preceding data bytes for Fragment 2 = 776 bytes.
5. **Fragment Offset for Fragment 2 = $776 / 8 = 97$.**

SOLVED EXAMPLE 2: CIDR Block Range & Usable Hosts

Problem: Analyze the CIDR block `192.168.10.64/26`. Find total addresses, subnet mask, usable host range, and broadcast address.

Solution:

1. Prefix $M = 26$ implies Host bits $H = 32 - 26 = 6$ bits.
2. Total Addresses = $2^6 = 64$ addresses.
3. Subnet Mask = 26 ones followed by 6 zeros = `255.255.255.192`.
4. Divisibility check: Binary of 64 = `01000000`, last 6 bits `000000` implies 64 is divisible by 64.
5. Network Address = `192.168.10.64`.
6. Broadcast Address = `192.168.10.127` ($64 + 63$).
7. **Usable Host Range: `192.168.10.65` to `192.168.10.126` (62 usable hosts).**

6.5 NPTEL Exam Practice Questions (MCQs)

Q1: In non-transparent fragmentation, where does reassembly of fragments take place?

- A) At the ingress router | B) At the egress router | C) Only at the destination host | D) At every hop router

EXAM POINT

Correct Answer: C | Explanation: Non-transparent fragmentation (IP standard) sends fragments independently; reassembly occurs solely at the final destination host.

Q2: What unit is used to measure the Fragment Offset field in an IP header?

- A) Bits | B) Bytes | C) 4-byte words | D) 8-byte multiples

EXAM POINT

Correct Answer: D | Explanation: Fragment offset specifies payload position in multiples of 8 bytes (64 bits).

Q3: Which flag bit in the IP header prevents a router from fragmenting a packet?

- A) MF Flag | B) DF Flag | C) SYN Flag | D) URG Flag

EXAM POINT

Correct Answer: B | Explanation: DF (Do Not Fragment) = 1 instructs routers to drop the packet if it exceeds MTU and return an ICMP error.

Q4: What is the default subnet mask for a Class B IP address?

- A) 255.0.0.0 | B) 255.255.0.0 | C) 255.255.255.0 | D) 255.255.255.255

EXAM POINT

Correct Answer: B | Explanation: Class B uses 16 network bits and 16 host bits, yielding natural mask 255.255.0.0 (/16).

Q5: Which IP address range is reserved for Class A Private IP addresses (RFC 1918)?

- A) 10.0.0.0 to 10.255.255.255 | B) 172.16.0.0 to 172.31.255.255 | C) 192.168.0.0 to 192.168.255.255 | D) 127.0.0.0 to 127.255.255.255

EXAM POINT

Correct Answer: A | Explanation: 10.0.0.0/8 is the Class A private address range.

Q6: Which transport protocol provides a connection-oriented, reliable byte-stream service?

- A) IP | B) UDP | C) TCP | D) ICMP

EXAM POINT

Correct Answer: C | Explanation: TCP provides connection-oriented, reliable, full-duplex byte-stream service.

Q7: What is the fixed header size of a User Datagram Protocol (UDP) packet?

A) 8 bytes | B) 20 bytes | C) 32 bytes | D) 60 bytes

EXAM POINT

Correct Answer: A | Explanation: UDP header is fixed at 8 bytes (Src Port, Dst Port, Length, Checksum).

Q8: Which TCP flag is used during the 3-Way Handshake to initiate connection synchronization?

A) FIN | B) RST | C) SYN | D) PSH

EXAM POINT

Correct Answer: C | Explanation: SYN (Synchronize) initiates connection establishment and synchronizes initial sequence numbers.

Q9: What TCP state is exploited during a SYN Flood Denial-of-Service (DoS) attack?

A) ESTABLISHED | B) HALF-OPEN (SYN_RCVD) | C) TIME_WAIT | D) CLOSE_WAIT

EXAM POINT

Correct Answer: B | Explanation: SYN Flood floods server with SYN packets, filling the connection queue in half-open SYN_RCVD state.

Q10: How many usable host IP addresses are available in a CIDR `/28` subnet block?

A) 16 | B) 14 | C) 30 | D) 62

EXAM POINT

Correct Answer: B | Explanation: Prefix /28 leaves $H = 32 - 28 = 4$ host bits. Total = $2^4 = 16$. Usable = $16 - 2 = 14$ hosts.

6.6 Viva Questions with Concise Answers

Q1: What is MTU and what happens when an IP packet exceeds it?

Answer: MTU (Maximum Transfer Unit) is the max frame payload a link can carry. If packet > MTU and DF=0, it is fragmented; if DF=1, it is dropped.

Q2: Why is reassembly done only at the destination host in IP?

Answer: To allow fragments to be routed independently across parallel paths and to prevent intermediate routers from wasting CPU and buffer memory.

Q3: What is the purpose of the Identification (ID) field in the IP header?

Answer: To group fragments coming from the same original IP datagram during reassembly.

Q4: Why must the Fragment Offset be a multiple of 8 bytes?

Answer: Because the offset field is 13 bits wide; scaling by 8 allows it to cover the full 65,535-byte IP size range.

Q5: What are the private IP address ranges defined in RFC 1918?

Answer: 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

Q6: What is the purpose of the loopback address 127.0.0.1?

Answer: It allows a host to send network traffic to its own internal software stack for testing and local process communication.

Q7: What is a 5-tuple TCP Association?

Answer: A 5-tuple uniquely identifying a TCP connection.

Q8: Differentiate between Well-Known ports and Ephemeral ports.

Answer: Well-known ports (0-1023) are static ports for server services; Ephemeral ports (>4095) are dynamic temporary client ports.

Q9: What are the three steps of the TCP 3-Way Handshake?

Answer: 1. Client sends SYN. 2. Server responds with SYN-ACK. 3. Client sends ACK.

Q10: What is a half-open TCP connection and how is it used in SYN Flood DoS?

Answer: A half-open state occurs when server sends SYN-ACK and waits for final ACK. In SYN Flood, fake SYNs exhaust the server's SYN queue memory.

6.7 Final Revision Checklist

- ☐ Understand MTU, Packet Fragmentation, DF/MF Flags, and 8-byte Offset calculations.
- ☐ Differentiate Transparent vs Non-Transparent Fragmentation.
- ☐ Identify Class A, B, C, D, E IP ranges, Network IDs, and Broadcast addresses.
- ☐ Memorize RFC 1918 Private IP ranges and Loopback address functionality.
- ☐ Compare TCP vs UDP characteristics, headers, and reliability mechanisms.
- ☐ Know key Well-Known ports (21, 22, 23, 25, 53, 69, 80, 443) and `/etc/services`.
- ☐ Explain TCP 3-Way Handshake, 4-Way Termination, and 5-Tuple Association.
- ☐ Understand Half-Open connections, SYN Flood DoS attack mechanics, and mitigations.
- ☐ Master Bitwise ANDing for Subnet IDs, VLSM design, and CIDR `/M` prefix calculations.